

Developing a cone-specific promoter using the *PDE6C* gene

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Background: Targeting cone photoreceptors for gene therapy requires promoters that drive specific expression within the limited packaging capacity of adeno-associated viral (AAV) vectors. However, commonly used cone-specific promoters such as the PR1.7 promoter are relatively large (~1700bp), which can restrict vector design when combined with large therapeutic genes of interest. In this study, we designed and evaluated novel promoters derived from the human *PDE6C* gene to identify shorter sequences capable of driving cone-specific expression.

Methods: Promoter sizes of 400bp, 700bp, and 1500bp were packaged into AAV7m8 vectors with an mCherry reporter gene. Constructs were administered to C57BL/6J mice and expression patterns were compared to established PR1.7 and ProA1 cone-specific promoters. Intravitreal injections were investigated as a less invasive approach for gene therapy, whilst subretinal injections were investigated as the gold-standard method for photoreceptor targeting.

Results: Following intravitreal administration, the PR1.7 promoter demonstrated the highest cone specificity, the *PDE6C* promoters transduce cones but also had some transduction of cells within the inner retina, whereas the ProA1 exhibited the least specificity and had broad expression across the retina. The *PDE6C* promoters showed a size-dependent increase in mCherry RNA expression, with the 1500bp construct having a higher expression than the PR1.7 promoter. When delivered subretinally, all *PDE6C*-derived promoters were capable of transducing cone photoreceptors to a high efficiency.

Conclusions: These findings demonstrate that human *PDE6C*-derived promoter sequences can drive transgene expression in the mouse retina. Although intravitreal delivery had lower specificity than the PR1.7 standard, all *PDE6C* promoter sizes transduced cone photoreceptors efficiently when delivered subretinally. These novel promoters provide a promising alternative for AAV-based gene therapies where current established promoters may exceed packaging capacity limits for large therapeutic genes.

